

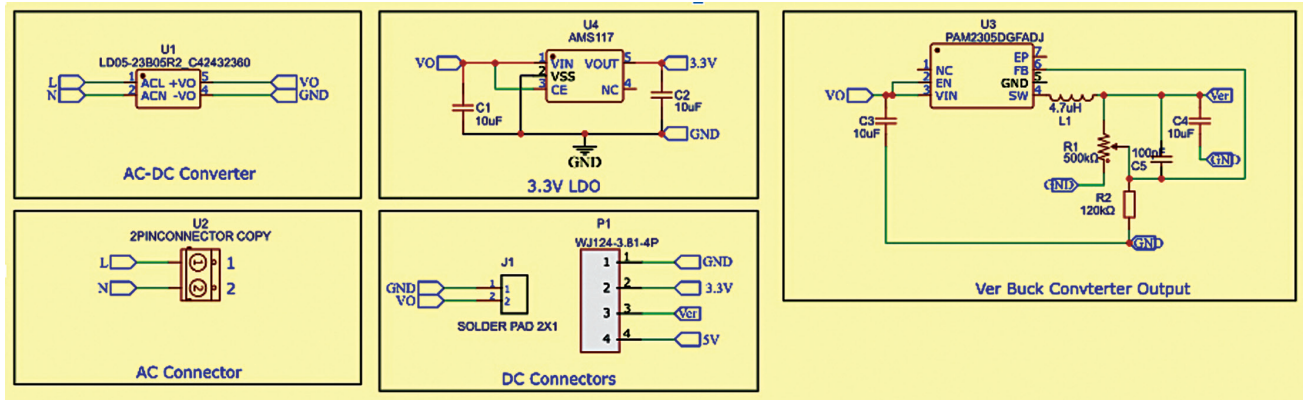


Fig. 5: Complete circuit of the AC-DC converter

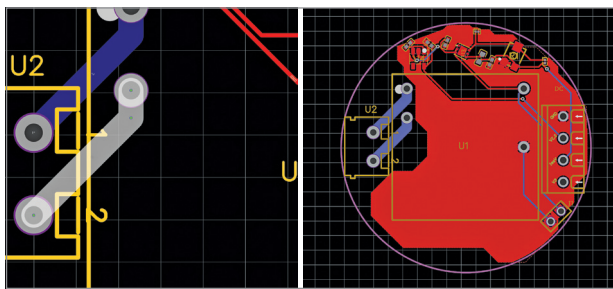


Fig. 6: PCB design showing the track width for the AC power line input

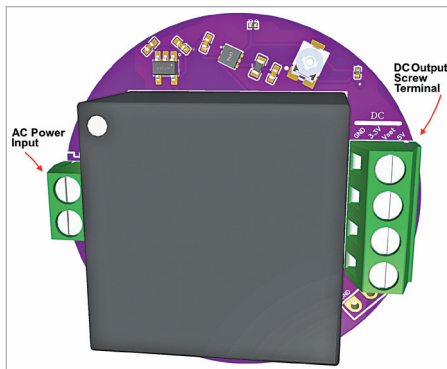


Fig. 7: Final PCB design

with a diameter of 3.8cm, designed for compactness and ease of assembly. Components are arranged in five logical zones:

1. AC-DC converter (U1). Positioned at the top left, with AC input terminals (L, N) and output (VO, GND). C1 and C2 are placed close for filtering.

2. Voltage regulator (U4). Located at the centre, receives input from U1 and supplies a 3.3V output,

stabilised by capacitors C3 and C4. This output is routed to connector J1.

3. Buck converter (U3).

Positioned at the top right, it includes components L1, R1, R2,

and C5, which together form the adjustable voltage output circuit. This output is also routed to J1 for selectable voltage access.

4. AC connector (U2). Placed at the bottom left, this connector receives the AC input, with L and N lines routed to U1.

5. DC output connector (J1). Located at the bottom right, it provides the 3.3V, 5V, and adjustable voltage outputs, along with GND terminals for reliable connections.

The AC input track width is kept sufficiently wide to handle mains voltage safely and is connected via a screw terminal block for secure wiring.

Testing

The module was tested using a multimeter, a variable AC power supply, a resistor load bank, an oscilloscope, and soldering tools. The testing procedure included

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verifying that the LD05-23B05SR2 (U1) outputs $5V \pm 5\%$ across a 100V-230V AC input range, the AMS117-3.3 (U4) provides $3.3V \pm 3\%$, and the PAM2305DGFADJ (U5) delivers adjustable voltages (e.g., 2.5V-4.0V) using the resistor network formed by R1 and R2. The module demonstrated an efficiency greater than 75%, a ripple of less than 100mV, and stable operation under load conditions of 0.5A through 1A. The WJ124-3.81-4P (J1) output connector was verified for reliability, with:

- Pin 1 providing 3.3V at maximum 800mA current
- Pin 2 providing 5V at maximum 1A current
- Pin 3 supplying an adjustable voltage at a maximum of 600mA current
- Pin 4 as GND **EFY**

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